



Recent advances of focused ultrasound induced blood-brain barrier opening for clinical applications of neurodegenerative diseases

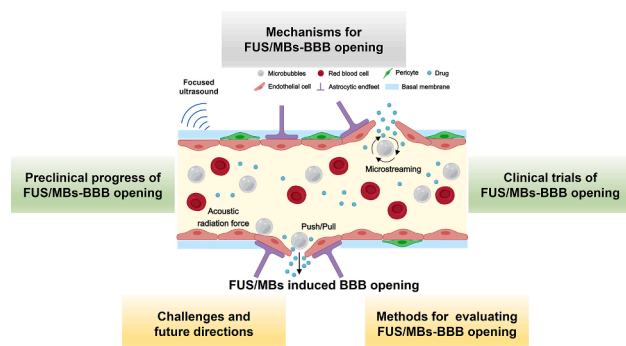
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GRAPHICAL ABSTRACT



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ABSTRACT

With the aging population on the rise, neurodegenerative disorders have taken center stage as a significant health concern. The blood–brain barrier (BBB) plays an important role to maintain the stability of central nervous system, yet it poses a formidable obstacle to delivering drugs for neurodegenerative disease therapy. Various methods have been devised to confront this challenge, each carrying its own set of limitations. One particularly promising noninvasive approach involves the utilization of focused ultrasound (FUS) combined with contrast agents—microbubbles (MBs) to achieve transient and reversible BBB opening. This review provides a comprehensive exploration of the fundamental mechanisms behind FUS/MBs-mediated BBB opening and spotlights recent breakthroughs in its application for neurodegenerative diseases. Furthermore, it addresses the current challenges and presents future perspectives in this field.

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1. Introduction

Neurodegenerative diseases include a broad range of clinical and pathological diverse conditions, such as Alzheimer disease (AD), Parkinson's disease (PD), amyotrophic lateral sclerosis (ALS), Huntington's disease, among others [1]. These diseases share a common feature: the loss of functionality in nerve cells within the brain, characterized by the misfolded proteins accumulating at the central nervous system (CNS). This leads to progressive functional impairment and, in severe cases, premature neuronal death [2]. It is well known that advancing age represent an enormous risk for the development of neurodegenerative diseases [3,4]. With the elderly population on the rise, the World Health Organization projects a threefold increase in the number of individuals afflicted by these diseases in the next three decades [5–7]. Unfortunately, effective clinical interventions for neurodegenerative diseases are still lacking. One of the foremost challenges in their treatment lies in the presence of the blood–brain barrier (BBB) [8,9]. The BBB is a complex structure composed of endothelial cells, tight junctions, basal membrane, neurons, pericytes, and astrocytic endfeet [10,11]. Both its chemical composition and physical attributes create a dynamic, adaptable interface crucial for maintaining the brain's system homeostasis. This plays a pivotal role in upholding the stability of the central nervous system's physiological environment, safeguarding it from potential harm caused by agents or microorganisms in the bloodstream [12–14]. Consequently, the BBB function poses a significant challenge for therapeutic agents aiming to penetrate into the CNS [12,15]. Studies indicate that molecules larger than 500 Da face a 100 % restriction, while over 98 % of small molecules encounter a similar barrier, impeding their passage through the BBB [16,17].

To bypass the BBB, several approaches have been developed [18,19]. Firstly, open surgery for direct injection, such as convection enhanced delivery (CED), which utilizes convection to supplement simple diffusion, thus enhancing the drug distribution in the brain [20–22]. Nevertheless, these invasive techniques come with surgical risks and extended recovery periods, rendering them unsuitable for repeated treatments [23]. Another approach involving the modification of targeting ligands has garnered significant attention [5,24]. However, its application is constrained by relatively low targeting efficiency [25]. Lastly, the intra-arterial injection of hyperosmotic solutions has been extensively studied for decades [26–29]. While this method is employed for BBB disruption in clinical settings, it necessitates general anesthesia, potentially leading to serious side effects [30,31].

Focused ultrasound (FUS), a non-invasive technique commonly used for imaging and diagnostics [10,13,16], is paired with microbubbles (MBs). These ultrasonic contrast agents consist of lipid or protein bubbles filled with inert gases like sulfur hexafluoride or perfluorocarbons. Notable Food and Drug Administration (FDA)-approved MBs include lipid-based Sonovue®, Definity®, and protein-based Optison® [32,33]. The combination of FUS and MBs (FUS/MBs) has generated significant interest for opening the BBB due to its non-invasive, safe, and repeatable performance [5,34]. In 2001, this method was first proved by Hynynen et al. to open BBB safely in rabbits. Since then, numerous laboratory and clinical studies focused on neurodegenerative diseases have utilized this method for BBB opening [35]. And in 2018, first clinical trial results of FUS/MBs BBB opening in AD patients was published [36].

This review will explore the mechanisms underlying BBB opening through FUS/MBs and methods for evaluating BBB permeability. It will provide an overview of both preclinical and clinical advancements using FUS/MBs-induced BBB opening in the neurodegenerative diseases. Additionally, it will discuss the challenges and future prospects in this evolving field.

2. Mechanisms involved in FUS/MBs-BBB opening

Combining FUS and MBs has presented a promising approach for opening the BBB, enhancing the accumulation of drugs in the brain. FUS

can penetrate the skull, forming a focused acoustic field, and alter the permeability of the brain. As the MBs circulate within the acoustic field, they oscillate and collapse, inducing localized cavitation. This confers non-invasive targeting capabilities. Furthermore, FUS/MBs can leverage cavitation by appropriately adjusting acoustic parameters such as frequency, amplitude, or waveform.

Over the past few years, this technique has demonstrated promising outcomes for malignancies and neurodegenerative diseases. Although extensive evidence from both preclinical and clinical studies confirms the effectiveness of BBB disruption through FUS/MBs, the precise mechanism underlying BBB opening remains elusive. Previous literature has identified three major characteristic effects induced by FUS: thermal, sonoporation, and cavitation [16,37–40]. The thermal effect entails an increase in tissue temperature following the absorption of FUS energy [41]. Sonoporation involves the transient formation of pores in cell membranes due to FUS irradiation, facilitating drug uptake [42,43]. Current consensus suggests that these two effects alone are not responsible for BBB opening, as there is no significant temperature elevation during FUS/MBs-mediated BBB opening [35]. Additionally, the pores created by the sonoporation effect are transient, whereas FUS/MBs-induced BBB opening can endure for several hours [37,43].

Hence, the focus narrows to the cavitation effect, which can be further categorized into stable and inertial cavitation based on how MBs collapse under ultrasonic waves [44–46]. At low mechanical indices, MBs exhibit periodic shrinkage and expansion, known as stable cavitation [47,48]. This phenomenon generates acoustic radiation force, push–pull mechanisms, and microstreaming, all of which contribute to BBB disruption [39]. Through acoustic radiation force, MBs move towards the vessel endothelium in the direction of ultrasonic wave propagation [49]. The push–pull mechanisms entail MBs' shrinkage pulling the vessel wall inward, while their expansion pushes the vessel wall outward, ultimately leading to the opening of tight junctions [50]. On the other hand, microstreaming induces shear stresses, which serve to further disrupt the BBB as shown in Fig. 1 [39,51]. When the mechanical index reaches a certain threshold, MBs collapse vigorously, giving rise to acoustic emissions, shockwaves, and micro-jetting, constituting inertial cavitation [51–54]. Some researchers posit that inertial cavitation is pivotal for BBB disruption [39,40]. Others contend that it may not be the primary factor, citing experimental results demonstrating BBB opening in the absence of acoustic emissions [37], which are characteristic of inertial cavitation [55–58].

3. Methods for evaluating FUS/MBs-BBB opening

The degree of BBB opening is crucial in FUS/MBs-induced procedures. Therefore, methods to ascertain the parameters of reciprocal biomechanical interactions between MBs and cerebral vessels, as well as real-time BBB opening volume and effective drug accumulation, are critical for further applications [59]. In many experiments of FUS/MBs-induced BBB opening, relatively uniform FUS parameters have been employed [16]. While this approach can yield relatively consistent results in thin-cranium or craniotomies applications, achieving personalized therapy remains challenging [60]. For instance, beam aberrations caused by thick craniums require adjustments of acoustic parameters based on computed tomography (CT) scanning and correction algorithms [61]. Different patients and diseases, particularly in the complex brain microenvironment, present challenges for individualized treatment. Techniques like magnetic resonance imaging (MRI), two-photon microscopy imaging, and other visualization tools can enhance the accuracy of evaluation.

MRI is capable of assessing the opening and restoration of the BBB, as well as detecting subtle lesions caused by FUS/MBs procedures such as edema and petechiae. However, patients often find it challenging to cope with the time and cost associated with repeated MRI scans. Additionally, the use of excessive contrast agents may lead to adverse events [62]. In recent years, the combination of fluorescence imaging

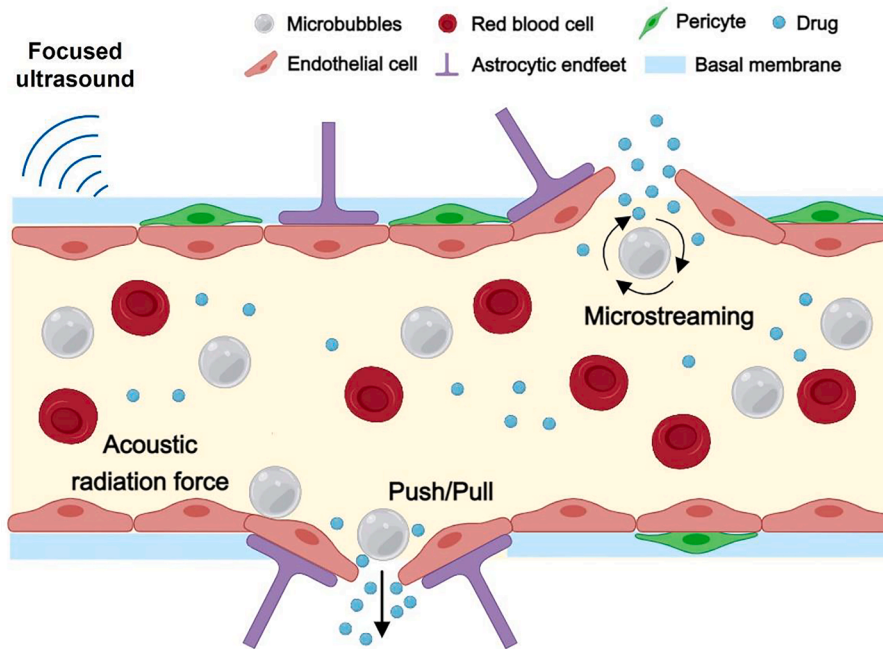


Fig. 1. Scheme of mechanisms involved in FUS/MBs-BBB opening. Three physical mechanisms (acoustic radiation force, push–pull mechanisms, and microstreaming), based on the stable cavitation effect, which might contribute to the opening of BBB. Figure created through [BioRender.com](#).

techniques with Evans blue (EB) dye has gained widespread use in studying vasculature leakage in the brains of animals [22,63]. Choi et al. used fluorescent dextrans with different molecular weights to assess the BBB opening induced by FUS/MBs, and found that dextrans with 3 or 70 kDa could cross the BBB, while dextran with 2000 kDa could not. Besides, higher concentration of 3 kDa dextran was crossed the BBB compared to the 70 kDa dextran [64]. Moreover, EB dye exhibits a strong binding ability to serum albumin, which normally cannot pass through the BBB. Consequently, when the BBB is compromised, EB-bound albumin may extravasate from cerebral vessels into adjacent tissues. While EB can evaluate BBB permeability, it falls short in assessing the spatial pattern of extravasation resulting from changes in BBB permeability. Furthermore, the heterogeneous distribution of macromolecules crossing the BBB also limits the precision and application of EB staining [65].

In addition, some researchers have utilized two-photon microscopy imaging in combination with dextran modified Texas Red to provide real-time observations of the cerebral vessels. By analyzing the time-dependent intensity changes on the extravascular region, BBB leakage was categorized as slow, sustained, and fast types. Slow leakage may result from transcytosis and paracellular routes, as well as fast leakage may stem from vessel wall punctures [66]. It's worth noting that the use of two-photon microscopy requires a craniotomy to access the skull, though it could achieve the visualization of brain blood vessels at the subcellular level *in vivo*. Therefore, there is a pressing need for more direct and real-time methods in future clinical applications.

4. Preclinical progress of FUS/MBs-BBB opening

4.1. FUS/MBs-BBB opening for Alzheimer disease therapy

Alzheimer's disease (AD) stands out as the most prevalent neurodegenerative disease, characterized by the widespread deposition of β -amyloid protein and neurofibrillary tangles throughout the brain, leading to a progressive decline in cognitive function [67]. Current treatments focus primarily on providing temporary relief of behavioral and psychological symptoms. As of now, there is no definitive cure for AD, and the BBB presents a significant challenge in achieving

therapeutic concentrations in the brain [23]. Preclinical experiments have demonstrated that the combination of FUS with MBs facilitates the passage of therapeutic agents across the BBB.

4.1.1. β -amyloid protein

Currently, most of the preclinical studies are focused on β -amyloid protein reduction. For example, Aubert et al. utilized intravenous immunoglobulin (IVIg), an immunomodulator known to reduce β -amyloid and enhance neurogenesis, in combination with MRI-guided focused ultrasound (MRIGFUS) and MBs to achieve a sufficient level of IVIg for AD treatment [68]. The increase in BBB permeability at the hippocampus was confirmed through T1-weighted contrast-enhanced MR images. With the assistance of FUS/MBs, the concentration of IVIg in the hippocampus reached 489 ng/mg after 4 h, significantly higher than in the control groups. Even after 2 weeks, nearly 20 ng/mg of IVIg was still detected in the hippocampus. To evaluate treatment efficacy, experiments were conducted on an AD mouse model, assessing β -amyloid plaques and neurogenesis. The FUS/MBs plus IVIg group exhibited the most notable reduction in the number and mean surface area of β -amyloid plaques, resulting in a 68 % decrease in plaque load compared to the saline group. Additionally, hippocampal neurogenesis was remarkably increased in the FUS/MBs plus IVIg group, reaching a level four times higher than in the saline group.

In another study, Götz et al. combined FUS/MBs and aducanumab (an anti- β -amyloid antibody which could reduce amyloid pathology) for AD treatment [69]. With the FUS/MBs treatment, the concentration of aducanumab in the brain lysate was measured at 21.77 ng/mL, as opposed to 4.32 ng/mL without FUS/MBs. This approach resulted in a robust improvement in spatial memory and a marked decrease in amyloid plaques in an AD mouse model.

In addition, Aubert et al. employed MRI-guided focused ultrasound (MRIGFUS) in conjunction with MBs to deliver D3, a selective tropomyosin receptor kinase A agonist, for AD therapy (Fig. 2) [70]. The successful opening of the BBB in an AD mouse model was confirmed by T1-weighted contrast-enhanced MR images. Further quantitative analysis revealed that, without FUS irradiation, the D3 level in the basal forebrain remained below the detection limit. In contrast, with the assistance of FUS/MBs, the D3 level in the basal forebrain reached $15 \pm$

MRlgFUS-mediated delivery of D3 to the basal forebrain

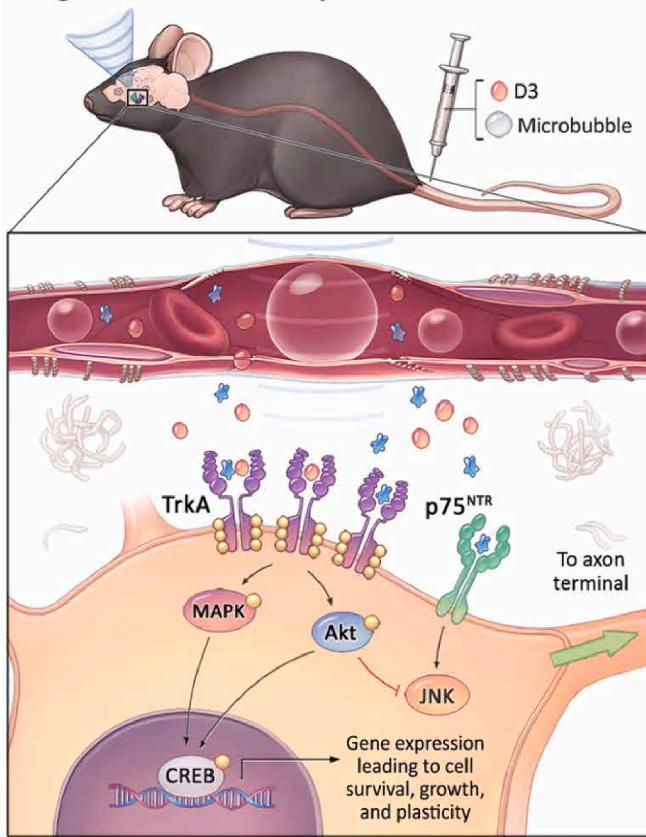


Fig. 2. The application of MRlgFUS in conjunction with MBs facilitated the delivery of D3 in an AD model. MRlgFUS was employed to enhance the permeability of the BBB, thereby increasing the bioavailability of D3. This led to the activation of TrkA, Akt, MAPK, and CREB, while concurrently inhibiting JNK. These molecular responses ultimately contributed to neuronal survival, growth, and plasticity. This information has been reproduced with permission. Copyright 2020, American Association for the Advancement of Science [70].

2 ng/mg of tissue in the AD mouse model. Encouraged by the increased brain delivery efficiency of D3, they proceeded to evaluate the cholinergic function in the AD mouse model. The *in vivo* data demonstrated that a single FUS/MBs plus D3 therapy could activate cholinergic function in the AD mice for up to 3 days.

4.1.2. Tau protein

Tau, a microtubule-associated protein, stands as another promising target for AD therapy [71]. Nisbet and colleagues conducted a comparison of the brain delivery efficiency of anti-tau antibodies in three formats: single-chain variable fragment, full-sized immunoglobulin G, and fragment antigen-binding, both with and without the application of FUS/MBs [72]. The brain distribution results demonstrated that all three antibody formats could traverse the BBB when combined with FUS and MBs. Among them, the full-sized immunoglobulin G antibody achieved the highest concentration in the brain compared to the other two antibody formats. Notably, it is worth mentioning that FUS/MBs alone, without the addition of any therapeutic agents, have shown therapeutic effects in some AD animal models [73,74].

4.2. FUS/MBs-BBB opening for Parkinson disease therapy

The second most common neurodegenerative disorder is Parkinson's disease (PD), characterized by the gradual loss of dopaminergic neurons and the overexpression of α -synuclein [23,75].

4.2.1. Neurotrophic factors

Current PD treatments primarily focus on dopamine replacement or mimicking its function [23]. Levodopa (L-dopa) serve as the first-line drug for PD therapy, capable of crossing the BBB and augmenting dopamine levels [76]. However, L-dopa does not halt the ongoing loss of dopaminergic neurons, which is associated with complications like motor fluctuations and dyskinesia [77,78]. Recent studies have revealed the potential of neurotrophic factors (NFs) in mitigating progressive neuronal loss, thereby supporting the preservation of the nervous system [79,80]. Nevertheless, these therapeutic factors face challenges in penetrating the BBB. In order to address this issue, Liu's group developed a complex consisting of liposomes and MBs for the delivery of a plasmid carrying both the green fluorescent protein gene and the glia-derived neurotrophic factor (GDNF) gene (named as the LpDNA-MBs complex, as shown in (Fig. 3) [81]. By utilizing FUS to induce BBB opening, the transfection efficiency of the plasmid increased by 5–10 times in the striatum compared to the control groups. This led to the liposome/MBs plus FUS group demonstrating the most significant improvement in dopaminergic neuronal function restoration and motor-related behavior recovery in a PD mouse model.

In a subsequent study, the same group further utilized the liposome/MBs complex to co-load with both the GDNF plasmid and brain-derived neurotrophic factor (BDNF) plasmid [82]. When combined with FUS exposure, this designed liposome/MBs complex successfully opened the BBB, enhancing the penetration and expression of the loaded plasmids in the brain. Evaluation results revealed that either the GDNF or BDNF plasmid delivered by the liposome/MBs complex could provide a neuroprotective effect, rescuing dopaminergic neuronal loss and promoting the recovery of motor-related behavior. However, the combination of the GDNF and BDNF plasmids did not exhibit an additional effect. This might be explained by the competing expressions between the GDNF and BDNF genes, which led to a reduction in the overall treatment effect.

4.2.2. α -Synuclein

Targeting α -synuclein is another potential therapy for PD [83]. Yan and co-workers developed nanobubbles loaded with curcumin, a polyphenol separated from the turmeric, which could efficiently remove α -synuclein [84]. When exposed to FUS irradiation, the BBB could be effectively opened, allowing curcumin to be delivered into the deeper regions of the brain by the nanobubbles. Behavioral tests conducted in a PD mouse model indicated that nanobubbles loaded with curcumin combined with FUS could effectively alleviate PD symptoms.

4.3. FUS/MBs-BBB opening for amyotrophic lateral sclerosis therapy

Amyotrophic lateral sclerosis (ALS) is a relatively rare yet devastating neurodegenerative condition, characterized by the degeneration of both upper (corticospinal motor neurons, CSMN) and lower (spinal motor neurons, SMN) motor neurons. This leads to muscular weakness, spasticity, paralysis, respiratory failure, and ultimately death [85]. Currently, only two drugs, Edaravone and Riluzole, have received FDA approval, though their benefits remain somewhat limited [86,87]. Furthermore, effectively delivering these drugs to specific regions of the brain with a sufficiently high concentration remains a considerable challenge due to the presence of the BBB.

4.3.1. Edaravone

Recently, Chen et al. conducted a study to examine the impact of Edaravone delivery to the motor cortex using FUS/MBs in a transgenic ALS mouse model (Fig. 4) [88]. T1-weighted contrast-enhanced MRI confirmed the successful BBB opening via FUS/MBs, leading to the accumulation of Edaravone in the motor cortices. Quantitative results revealed that the concentration of Edaravone in the motor cortices was 117.7 ± 27.3 ng/g for the FUS/MB plus Edaravone group, whereas it was only 52.6 ± 16.5 ng/g without FUS. Motor functions of the mice were assessed through wire hanging tests, grip

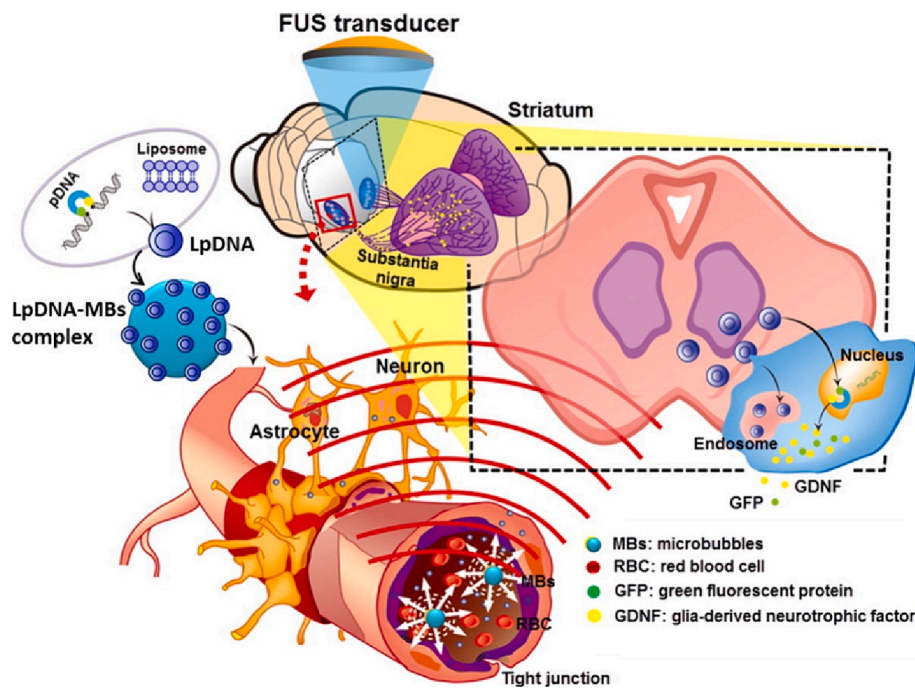


Fig. 3. Schematic illustration of the liposome/MBs complex facilitating FUS induced BBB opening for PD therapy. Exposed to FUS irradiation, the designed LpDNA-MBs complex opened the BBB and eventually entered the striatum cells, thus increasing the protein expressions, and playing therapy effect in PD mouse model. Reproduced with permission. Copyright 2016, Elsevier [81].

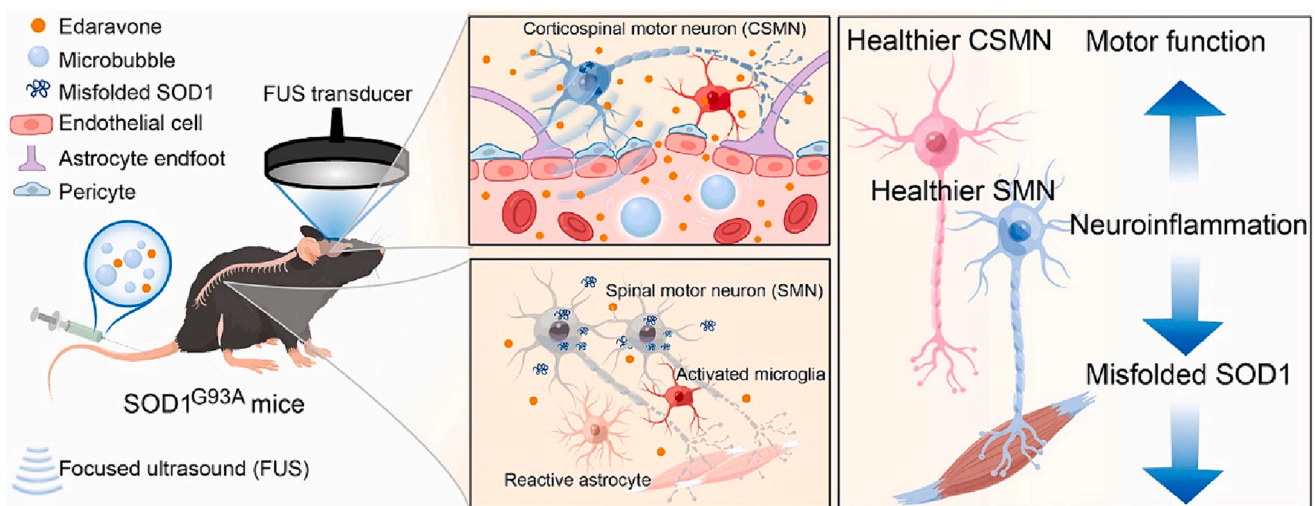


Fig. 4. Schematic illustration for the combination of FUS/MBs and Edaravone to open the BBB and ameliorate ALS. Upon FUS exposure, the BBB was disrupted by the cavitation effect of MBs, which increased the Edaravone delivery to the CSMN and SMN. Through activating the microglia and astrocyte cells, the health of CSMN and SMN were improved due the increased motor function, as well as decreased neuroinflammation and misfolded SOD1 protein. Reproduced with Permission. Copyright 2023, Elsevier [88].

strength measurements, and gait analysis, showing significant motor function impairments in the FUS/MB plus Edaravone group. Immunostaining data demonstrated remarkable neuroprotection, microglia recovery, astrocytosis improvement, as well as reduction of misfolded SOD1 protein and neuroinflammation in the FUS/MB plus Edaravone group in the transgenic ALS mouse model.

4.3.2. Neuroprotective agents

Recent preclinical studies suggest that combining neuroprotective and neurogenerative strategies with FUS/MBs-induced BBB opening may offer a promising approach. Shen et al. reported notable improvements in BBB permeability for neuroprotective agents (arctiin) through

the application of FUS in conjunction with MBs [89]. Additionally, this intervention led to enhanced neuromuscular function in mouse models.

4.4. FUS/MBs-BBB opening for Huntington's disease therapy

Huntington's disease (HD) is an inherited neurodegenerative condition characterized by the gradual onset of involuntary choreiform movements, alongside behavioral and cognitive impairments [90]. Presently, therapy for HD primarily focuses on providing symptomatic relief, as no disease-modifying treatment has been identified [91,92]. While certain potential drugs have shown promising therapeutic effects in preclinical studies, including dopamine stabilizers,

immunomodulators, and NFs [93–96]. For instance, Liu et al. constructed a liposome loaded with a plasmid encoding GDNF (named as GDNFp-LPs) to explore the feasibility of combining FUS/MBs for HD gene treatment (Fig. 5) [95]. Coupled with FUS and MBs to facilitate BBB permeation, the delivery of GDNF was significantly enhanced within the HD-afflicted brain of transgenic mouse model. The results of immunostaining, MRI and rotarod test from HD mice demonstrated that GDNF overexpression led to the decreased aggregation of polyglutamine-expanded aggregates, the reduction of oxidative stress and inflammation, as well as an improved neuroprotective effect and motor performance.

Table 1 has summarized the preclinical progress of FUS/MBs-mediated BBB opening in neurodegenerative diseases, including AD, PD, ALS and HD. Overall, plenty of preclinical studies have proved that the utilization of FUS/MBs-induced BBB opening is an effective supplement for neurodegenerative diseases therapy. Because it could enhance the brain delivery of different therapeutic agents in animal models, thereby improving the treatment effect of neurodegenerative diseases.

4.5. Other applications

In addition to its application in BBB opening, FUS also finds utility in the treatment of ischemic stroke, with or without the use of MBs, a technique known as sonothrombolysis [97–99]. A preclinical study by Hölscher's group successfully demonstrated the effectiveness of sonothrombolysis through the application of FUS in conjunction with MBs in a rabbit stroke model [100]. Furthermore, FUS alone can be employed for the control of epilepsy through neuromodulation [101–103].

5. FUS/MBs-BBB opening in clinical trials

The clinical application of FUS combined with MBs for BBB opening has generated substantial interest, especially given the encouraging results observed in preclinical studies. Table 2 provides an overview of clinical trials investigating FUS/MBs-mediated BBB opening in neurodegenerative diseases, encompassing AD, PD, and ALS.

5.1. FUS/MBs-BBB opening in clinical trials for Alzheimer disease

In the initial published clinical trial (NCT02986932), Lipsman and colleagues conducted an open-label, prospective, proof-of-concept, phase I study to explore the feasibility and safety of FUS/MBs-mediated BBB opening in early to moderate AD patients [36]. The study enrolled 5 patients (three males and two females with an average age of 66.2 years) in the mild-to-moderate stage. The results demonstrated that the BBB was successfully opened in the frontal lobe of all patients treated with FUS/MBs, and the BBB closure was observed after 24 h. No patient experienced any serious adverse events, including swelling, neurologic deficit, hemorrhage and death during the study. Moreover, no clinically meaningful changes were detected in patient cognition, daily functioning, or β -amyloid levels. It's important to note that this represents the first clinical study utilizing FUS/MBs for BBB opening in human AD patients; however, it does have some limitations. Firstly, the sample size was small, which may limit generalizability. Secondly, the study was not primarily designed to assess efficacy, so no definitive conclusions can be drawn regarding the impact of FUS on the clinical symptoms of AD patients.

Similarly, based on an ongoing open-label phase II clinical trial (NCT03671889),

Rezaei and his team enrolled six participants (five female and one male, ages from 55 years to 73 years) with early AD [104]. FUS irradiation was performed in the hippocampus/entorhinal cortex. The results demonstrated that all participants tolerated the FUS procedure well, without experiencing any neurological changes or therapy-related adverse effects. Notably, there were no clinically significant changes observed in the formal cognitive assessments. Furthermore, the researchers speculated that the extent of Blood-Brain Barrier (BBB) opening corresponded to a range of 14 % to 71 % (with an average of 29 %) of the entire hippocampus volume. This variation was attributed to factors such as the number of targets, dosing, anatomical differences, and hippocampal atrophy. In a subsequent study, Mehta's team enrolled three female patients (aged 61, 72, and 73, respectively) with early AD [105]. The main limitations of this study were small sample size and contained only female participants. Overall, these two studies presented

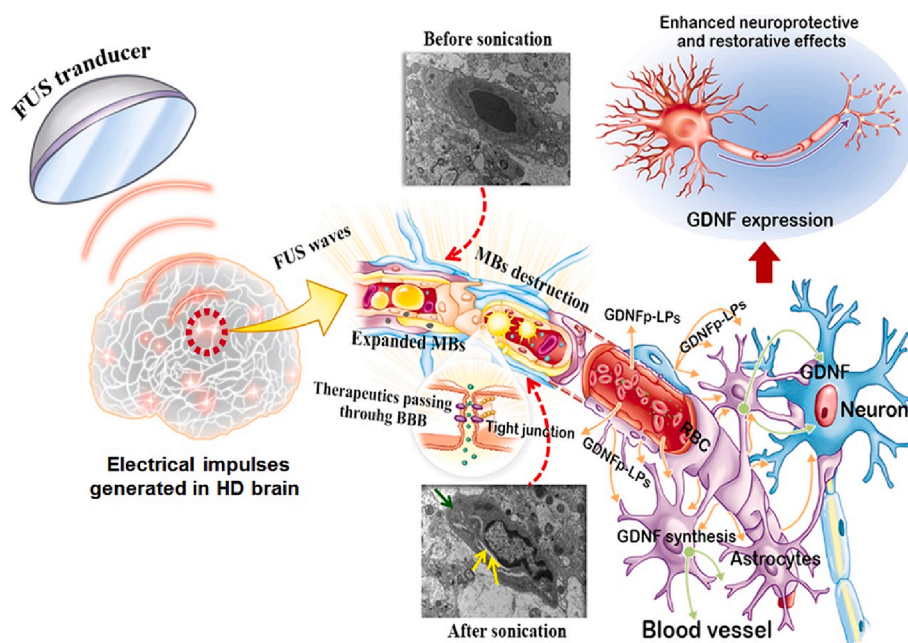


Fig. 5. FUS/MBs-mediated BBB opening facilitating delivery of GDNF plasmid-liposomes (GDNFp-LPs). The small green circles depict GDNFp-LPs, while the yellow circles represent MBs. After FUS exposure, the MBs reversibly destructed the BBB, and the prepared GDNFp-LPs entered into neuron cells, which increased the expression of GDNF. Ultimately, the neuroprotective effect was induced due to the overexpression of GDNF. Reproduced with Permission. Copyright 2019, Elsevier [95]. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1
Preclinical progress of FUS/MBs-mediated BBB opening in neurodegenerative diseases.

Disease model	MBs type	Delivered agents	Main results	Reference
<i>Alzheimer disease</i>	Definity®	Intravenous immunoglobulin (IVIg)	IVIg combined with FUS/MBs-mediated BBB opening, can deliver IVIg to the hippocampus for plaque reduction and neurogenesis promotion.	[68]
	Phospholipid shell, octafluoropropane gas core	Aducanumab	Aducanumab across the BBB through FUS/MBs significantly reduced the amyloid plaque.	[69]
	Definity®	Selective TrkA agonist (D3)	FUS/MBs-induced delivery of D3 localized to the basal forebrain, and was able to rescue the cholinergic function.	[70]
	Phospholipid shell, octafluoropropane gas core	Anti-tau antibody with three formats	All three antibody formats were able to across the BBB with FUS/MBs, and the full-sized immunoglobulin G antibody reached the highest concentration in the brain.	[72]
<i>Parkinson disease</i>	Avidinylated MBs conjugated with liposome	Liposome containing GDNF plasmid	FUS irradiation could induce BBB opening, and increase the plasmid transfection, thus recovering dopaminergic neuronal function and motor-related behavior.	[81]
	Phospholipid shell, sulfur hexafluoride gas core	GDNF and BDNF plasmid	FUS exposure could improve the brain delivery of plasmids, but the two plasmids did not exhibit an additional effect.	[82]
	Lipid-PLGA nanobubbles	Curcumin	Nanobubbles loaded curcumin combined with FUS were able to open the BBB and deliver the curcumin into the brain, thus increasing the efficacy of curcumin.	[84]
<i>Amyotrophic lateral sclerosis</i>	Phospholipid shell, perfluoropropane gas core	Edaravone	FUS/MBs increased the brain delivery of Edaravone, and provides additive amelioration on ALS mouse model.	[88]
	Phospholipid shell, perfluoropropane gas core	Arctiin	Improved delivery of arctiin through FUS/MBs showed treatment effect on ALS mouse model.	[89]
<i>Huntington's disease</i>	Sonovue®	Liposome containing GDNF plasmid	FUS/MBs facilitated delivery of liposome-GDNF, and enhanced motor performance in mouse model.	[95]

findings affirming the safe opening of the BBB in the hippocampus and entorhinal cortex of early AD patients [104,105]. Additionally, D'Haese et al. reported that FUS/MBs-induced BBB opening was not only well-tolerated but also resulted in a reduction of β -amyloid plaques (averaging 5.05 %) [106]. Unfortunately, despite these promising outcomes, no significant clinical improvement was observed in the formal cognitive assessments.

Given the extensive neuropathological changes associated with AD, it is imperative to evaluate the feasibility of opening the BBB over a larger area. Building upon a single-arm, open-label study (NCT03739905), Lipsman et al. enrolled nine patients with mild-to-moderate AD (five females and four males, with a mean age of 70 years) [107]. The volume of tissue treated with FUS was notably large, averaging 9.2 cm³ (ranging from 2.8 to 15.7 cm³), and MRI results confirmed the successful BBB opening post FUS irradiation. Additionally, no new neurological deficits or significant adverse events including headaches confusion, redness and oedema were observed by clinical examination. Except to the small sample size, other limitations included the absence of control and placebo arms. Similarly, Chang et al. designed an open-label, prospective study (NCT04526262) to investigate the feasibility, efficacy, and safety of extensive and repeated BBB opening in the frontal lobe of moderate-to-severe AD patients [108].

They enrolled and analyzed four female and one male patients with AD (with a mean age of 70.6 years) on the moderate to severe stage. All the five participants were well tolerable during the FUS procedures without any adverse events including overt cerebral hemorrhage, brain edema, infarction, or systemic and neurological deterioration. Notably, FUS targeted an extensive volume of the bilateral frontal lobes (21.1 ± 2.7 cm³), with the BBB confirmed to be opened at 95.7 ± 9.4 % of the targeted volume. Additionally, a slight reduction in β -amyloid deposition was observed in the frontal-to-other cortical region following FUS/MBs-induced BBB opening, although no significant cognitive improvements were noted. The main limitations of this study can be summarized as a small sample size, the absence of a placebo-controlled arm, and the lack of objective cognitive improvements. In summary, both of these clinical trials, NCT03739905 and NCT04526262, have demonstrated that the combination of FUS with MBs can indeed achieve BBB opening over a substantial area [107,108].

Encouraged by the safety and feasibility demonstrated by previous studies, a new trial (NCT05469009) is currently underway to assess the safety, feasibility, and potential clinical impact of integrating standard Aducanumab therapy (a human monoclonal antibody approved for AD

treatment [109]) with FUS/MBs-induced BBB opening in patients diagnosed with mild AD.

5.2. FUS/MBs-BBB opening in clinical trials for Parkinson disease

Building upon a proof-of-concept, single-arm, prospective, non-randomized phase I clinical trial (NCT03608553), Obeso et al. initially reported that BBB opening using FUS/MBs in the parieto-occipito-temporal junction of patients with PD was both safe feasible and reversible [110]. In this study, five male patients (average age of 73.2 years) with a mean PD duration of 7.6 years were recruited. The BBB opening in four of the five patients was successfully performed after FUS/MBs treatment without serious adverse events such as edema and hemorrhage. In their initial study, Obeso et al. observed minor adverse effects, including local phlebitis, anxiety, raised blood pressure, and confusion, though these resolved swiftly. Interestingly, they also noted mild cognitive improvements through various cognitive tests, with no significant changes detected in amyloid levels. In a subsequent follow-up study, they further affirmed the safety and feasibility of FUS/MBs-mediated BBB opening, this time targeting the posterior putamen [111]. They recruited seven male patients with PD, with an average age of 69.1 years. The results revealed that the BBB was safely and successfully opened in the posterior putamen region for all enrolled patients. The closure of the BBB occurred promptly after FUS procedures (24 h to 14 days). No serious adverse events, such as intracerebral hemorrhage or infarction, were recorded throughout the study. Although mild vasogenic edema occurred, it completely resolved within three months. Additionally, they found a reduction of β -amyloid on the targeted putamen region, while no significant changes were observed in cognitive assessments. Collectively, these two studies demonstrate the potential of FUS/MBs as an effective tool for BBB opening, promising advancements in the delivery of PD drugs.

As a result, there is a current emphasis on active clinical trials that are exploring the combination of FUS/MBs and PD drugs. Two ongoing trials, one being a proof-of-concept, open-label, prospective phase I trial (NCT04370665) and the other a two-arm, open-label phase I/II trial (NCT05565443), are focused on assessing the safety and feasibility of FUS/MBs-induced BBB opening for Cerezyme® delivery—an analogue of the human enzyme beta-glucocerebrosidase, showing promise for PD therapy [112,113]. According to the information available from NCT04370665, Kalia et al. enrolled four PD patients (two females and two males) with an average age of 54 years [114]. The results

Table 2

Clinical trials of FUS/MBs-mediated BBB opening in neurodegenerative diseases.

Disease	Study ID	MBs/Drug	Outcome measures	Current results	Status	Start
Alzheimer disease	NCT02986932	Definity®	1. Efficacy of BBB opening 2. Change in amyloid uptake 3. Adverse events 4. Change in cognitive function	FUS/MBs induced BBB opening was safe, reversible and repeated in the frontal lobe of early AD patients [36].	Completed	2016
	NCT03671889	Definity®	1. Adverse events 2. Efficacy of BBB opening	The BBB opening was safe, transient, reproducible and noninvasive by FUS/MBs in the hippocampus/entorhinal cortex of early AD patients [104]. FUS/MBs induced BBB opening was safely performed in the hippocampus/ entorhinal cortex of early AD patients without any adverse effects [105]. FUS/MBs could not only induce BBB opening, but also decrease the β -amyloid plaque with an average reduction of 5.05 % in early AD participants [106].	Recruiting	2018
	NCT03739905	Definity®	1. Adverse events 2. Efficacy of BBB opening 3. Change in cognitive function 4. Change in Amyloid Tracer uptake	Large, multi-volume brain regions can be safely opened by FUS/MBs in mild-to-moderate AD patients [107].	Recruiting	2018
	NCT04526262	Definity®	1. Adverse events 2. Efficacy of BBB opening	FUS/MBs can extensively, safely and repeatedly induce BBB opening in the frontal lobe of moderate-to-severe AD patients. In addition, the BBB opening might be beneficial for amyloid removal [108].	Recruiting	2020
	NCT04118764	Definity®	1. Number of individuals with successful BBB opening 2. Number of safety events related to BBB opening 3. Change in amyloid uptake 4. Change in cognitive function	Not available	Recruiting	2020
Parkinson disease	NCT05469009	Definity®/ Aducanumab	1. Adverse events 2. Change in amyloid uptake 3. Change in cognitive function	Not available	Recruiting	2022
	NCT03608553	Luminy®	1. Adverse events	FUS/MBs induced BBB opening was safe, reversible in the parieto-occipito-temporal junction of PD patients [110]. FUS/MBs induced BBB opening was feasible and safe in the posterior putamen of PD patients [111].	Recruiting	2018
	NCT04370665	Definity®/ Cerezyme®	1. Efficacy of BBB opening 2. Adverse events 3. Effectiveness of Cerezyme® delivery 4. Effectiveness of Cerezyme® treatment	FUS/MBs induced BBB opening was feasible and safe for facilitating Cerezyme® delivery in PD patients [114].	Recruiting	2020
	NCT05565443	Definity®/ Cerezyme®	1. Adverse events 2. Efficacy of BBB opening	Not available	Not yet recruiting	2022
Amyotrophic lateral sclerosis	NCT03321487	Definity®	1. Adverse events 2. Efficacy of BBB opening	The reversible BBB opening using FUS/MBs was safe and tolerated in ALS patients [115].	Recruiting	2018

demonstrated that the BBB could be safely and reversibly opened using FUS/MBs in all four patients. Adverse events, including near vasovagal reactions, headaches, dyskinesia, edema, and ischemic complications, were of mild to moderate severity and transient in duration. It is noteworthy that glucose metabolism in the targeted putamen region showed a reduction in all PD patients after the combination of FUS/MBs and Cerezyme®. Specifically, the reduction in glucose metabolism was from 1.66 ± 0.14 to 1.27 ± 0.08 in the treated putamen region, and from 1.66 ± 0.14 to 1.64 ± 0.10 in the contralateral putamen region. Furthermore, glucose metabolism did not exhibit any significant changes in the untreated regions, including the substantia nigra and cortical regions. This provides evidence that FUS/MBs can enhance the brain delivery of

Cerezyme®. The limitations of this study mainly revolved around the small number of patients and the potential effects of a placebo. As for the clinical trial NCT05565443, it is currently ongoing, and no results have been posted as of now.

5.3. FUS/MBs-BBB opening in clinical trials for amyotrophic lateral sclerosis

While the majority of current clinical trials have been primarily focused on AD and PD, there is one ongoing trial specifically addressing Amyotrophic Lateral Sclerosis (ALS) (NCT03321487). This is a prospective, single-arm, open-label, first-in-human trial designed to assess

the feasibility, tolerability, and safety of BBB opening using FUS in conjunction with MBs for patients with ALS. In a recent study published by Abrahao et al., two female and two male ALS patients with an average age of 61 years were enrolled [115]. Outcomes have demonstrated that FUS/MBs-induced opening of the BBB was successful and well tolerated in the primary motor cortex of all four participants. There were no serious adverse events such as ischemia, hemorrhage, or gliosis, and only a few mild-to-moderate adverse events, including headache, vagal response, scalp pain, edema, bruising, musculoskeletal pain, and scalp petechial rash, were reported. Additionally, no significant clinical changes were observed in cognition and spasticity measurement. Although this study has some limitations, such as a small sample size and challenges in accurately and quantitatively evaluating BBB opening, as well as difficulties in identifying the functional areas of the primary motor cortex, it is noteworthy that this study represents a significant milestone in proving that non-invasive BBB opening in the motor cortex of ALS subjects by FUS/MBs is indeed feasible, reversible, and safe.

6. Challenges and future directions

The rapid progress in therapeutic applications made by FUS/MBs has revealed encouraging potential over the past decade. While inspiring examples of FUS/MBs continuously show improvements in managing various diseases, the persistent challenge of lacking cellular and molecular specificity remains not comprehensive enough for clinical application. In order to further benefit patients, there are still certain challenges that need to be discussed:

First, while recent clinical data have indeed affirmed the relative safety of combining FUS combined with MBs for BBB opening in patients with neurodegenerative diseases, it is crucial to acknowledge the potential for serious adverse events, particularly when employing more aggressive therapeutic approaches [116–119]. Notably, among the various biological effects induced by FUS, it is important to recognize that the sterile inflammatory response carries a probability of triggering a quick release of damage-associated molecular patterns [120]. The resulting pro-inflammatory cytokines within the treatment scenario hold the potential to create a complex and unbalanced immune micro-environment. Understanding the ramifications of this situation in the context of treating neurodegenerative diseases is challenging [121,122]. Previous studies have proved that FUS/MBs might lead to brain injury, encompassing intracerebral hemorrhage, ischemia, transient tissue edema, aseptic inflammation, and astroglial scarring [120,123,124]. Consequently, it is imperative to undertake comprehensive investigations into the long-term safety of recurrent BBB opening, involving a substantial sample size of patients.

Second, the precise mechanism underlying FUS combined with MBs induced BBB opening remains elusive, impeding our ability to accurately explain and predict post-BBB opening phenomena. Additionally, there is a shortage of comprehensive studies regarding the influence of FUS parameters. While previous literature has offered partial insights into the effects of parameters such as pressure amplitude, frequency, burst length, and sonication duration, establishing standardized criteria for optimal parameter selection remains a complex challenge [33,37,125–131]. In future research endeavors, the utilization of mathematical simulations and experimental validation tailored to specific neurodegenerative diseases may provide a pathway to refining FUS parameters. Furthermore, with an enhanced understanding of the mechanisms governing FUS/MBs-induced BBB opening, it may become possible to optimize the process while minimizing adverse effects.

Third, MBs are key elements for FUS mediated BBB opening, however, lack of special design and development for the BBB opening. In most of the correlational studies, MBs have directly used the commercially available US contrast agents [37]. In fact, the characteristics of MBs, including size, shell composition and gas core, could affect the FUS induced BBB opening [132–134]. Meanwhile, these characteristics of MBs are quite different in the commercially contrast agents. For MBs

size, several studies have shown that larger MBs could lead to higher degree of BBB opening and longer recovery time [135–137]. For shell composition, McDannold et al. reported that albumin-shelled MBs produced a higher degree of BBB opening compared with phospholipid-shelled MBs [33,138]. For gas core, Omata et al. demonstrated that the MBs contained perfluoropropane or perfluorobutane could open the BBB more efficiently than the MBs contained sulfur hexafluoride [139]. In short, next step research need to find out how the properties of MBs affect the FUS mediated BBB opening, thus optimizing the MBs for this application.

Fourth, current clinical trials primarily focus on validating the feasibility and safety of combining FUS with MBs for BBB opening, with only a small subset incorporating therapeutic drugs. The strategy of drug loading based on delivery systems has made significant strides in disease treatment. In addition to assisting FUS-BBB opening, MBs themselves can also deliver therapeutic agents and exert effects beyond vascular confinement [140,141]. Furthermore, by modifying the surfaces of drug-loaded MBs, selective drug delivery—targeting specific cell phenotypes—can be achieved through the addition of specific ligands such as antibodies, peptides, and aptamers [142,143]. These chemical conjugation technologies have matured significantly in MB design and preparation. Additionally, the combination of FUS/MBs with nanoparticles loaded with drugs is an appealing approach due to the various excellent performances of nano drug delivery systems [144,145]. This strategy holds the potential for widespread applications in advancing diagnostic and therapeutic agents for neurodegenerative diseases. Therefore, the development of sonication protocols involving the adjustment of parameters to realize both BBB permeability improvement and effective drug enrichment are critical for applications. Different types of drugs, including nucleic acid drugs, chemotherapeutic drugs, and protein drugs, possess distinct physiological characteristics when crossing the BBB. Consequently, the behavioral study of different drugs in the FUS/MBs treatment is eagerly anticipated. In the subsequent phase of research, it becomes imperative to establish methods for quantifying the extent of drug passage through the BBB. Moreover, conducting quantitative comparisons of drug penetration across the BBB between FUS/MBs and other BBB-crossing technologies holds significant scientific value [22].

Lastly, despite significant progress, there is still a need for more advanced instruments and methods to accurately assess BBB opening and establish the correlation between the cavitation signals and the level of BBB opening. Numerous studies have concentrated on monitoring the BBB opening in real time [146,147]. To ensure relevant bioeffects and enhance safety, scientists have exerted substantial efforts to ascertain the correct acoustic parameters for personalized therapy. Two types of cavitation detection—Passive Cavitation Detection (PCD) and Active Cavitation Detection (ACD)—are integral to FUS. PCD directly detects the activity of MBs during FUS exposure by quantifying the acoustic emission signals [148].

Real-time optimization of FUS/MBs-induced BBB opening through PCD typically comes by the cost of resolution, as the acoustic wave directly affects resolution in conventional reconstruction technology [149]. ACD, on the other hand, employs a secondary low-pressure pulse to evaluate the alterations created by initial pulse. It is accomplished using ultrasound scanning systems mapping the cavitation domain through capturing the backscattered signals of MBs [150]. However, a drawback of ACD is that MBs could only be indirectly monitored following the extended pulse processing to maintain high accuracy and resolution in the axial direction. Although the measurement pulse in ACD has considerably less energy than the primary pulse, it still might influence the cavitation process. This restricts the potential use for therapy feedbacks and parameters optimization throughout the BBB opening induced by FUS/MBs [151]. The methods outlined above can be employed to some degree for achieving real-time monitoring of BBB opening in FUS/MBs.

In summary, while current clinical research on this novel technology

may not be extensive enough for a comprehensive understanding, we anticipate that FUS/MBs will increasingly emerge as an attractive modality for combination treatment in neurodegenerative diseases.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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